

Problem Statement

Healthcare information is often scattered across large medical documents, treatment guidelines, and medicine reference manuals. Healthcare professionals, students, and general users may find it difficult and time-consuming to search through hundreds of pages to locate accurate information about diseases, medicines, dosages, and treatment procedures.

Traditional chatbots can provide answers, but they may generate inaccurate information (hallucinations) because they do not have access to specific medical documents. This can reduce the reliability and trustworthiness of the responses.

To address this problem, there is a need for an intelligent healthcare question-answering system that can retrieve relevant information from trusted medical documents and generate context-aware responses. The proposed Healthcare RAG (Retrieval-Augmented Generation) Assistant uses medical PDF documents as a knowledge base, converts them into vector embeddings, stores them in a Pinecone vector database, and retrieves the most relevant information based on user queries. The retrieved information is then combined with a Large Language Model (Gemini) to generate accurate and context-aware answers.

The objective of this project is to develop a Healthcare RAG Assistant that enables users to ask healthcare and medicine-related questions in natural language and receive reliable answers based on medical documents, thereby improving information accessibility, reducing search time, and minimizing the risk of misinformation.